

Local Subsidies Fraud Risk - Model Comparison

Note: Korean narrative is in docs/metrics_guide.md. This PDF summarizes aggregate metrics only (no raw rows).

Train period: 202401-202512

Test period: 202507-202512

Risk score range: 0-1000 (higher = higher fraud risk).

Reports contain aggregate statistics only. No raw PII rows.

See docs/metrics_guide.md for Korean explanations of metrics.

Per-algorithm reports: outputs/reports/{algorithm}/

Model metrics (aggregate)

algorithm=catboost_v3, Accuracy=0.9950607324423879, Precision=0.3562386980108499, Recall=0.7198538367844093, F1=0.47661290322580646, ROC_AUC=0.9556160903361884

algorithm=stacked_ensemble_v3, Accuracy=0.968785198940607, Precision=0.07516114180478821, Recall=0.7953714981729598, F1=0.13734356925018404, ROC_AUC=0.9520495060329013

algorithm=easy_ensemble_v3, Accuracy=0.7586608420347651, Precision=0.00886538733112614, Recall=0.6881851400730816, F1=0.017505267071508242, ROC_AUC=0.790329608546875

algorithm=gradient_boosting_v3, Accuracy=0.9866967030959847, Precision=0.1532278973295307, Recall=0.7198538367844093, F1=0.2526720820863617, ROC_AUC=0.947755332850089

algorithm=random_forest_v3, Accuracy=0.9976939937288807, Precision=0.8267477203647416, Recall=0.3313032886723508, F1=0.47304347826086957, ROC_AUC=0.9465353509505758